

LIMEN Connectome — Node Regulatory Reference

(123 canonical set)

Compiled: July 8, 2026 · **For:** `brain-node-domains.json` / `brain-connectome.json`
regulatory-dynamics layer **Scope:** Definition + self-regulation (autoregulatory mechanism) +
regulation-of-others (efferent modulation, with sign where known) for every entry in the
VERIFIED canonical 123-node set.

Read first — two limits on this document.

1. **Regulatory relationships are literature-standard neuroanatomy**, not re-verified DOIs this session. Where a relationship was DOI-verified in this session's citation work, it's marked ✓cited. Everything else is consensus textbook neuroanatomy — solid, but if you want each mechanism DOI-locked that's a separate verification pass.
2. **"Regulates others" is given from the literature, not from your L2 edge file.** I don't have `brain-connectome.json`'s 65 edges here. This document says what the edges *should* be per neuroscience; reconciling that against the 65 actually encoded edges (and their signs) is the next pass — and it's exactly the spec you need for the inhibition-first fix.

AUDIT VERDICT (correctness check)

- **Canonical count: 123 — CONFIRMED.** Counted directly.
- **SN collision: CONFIRMED, serious.** Canonical set disambiguates (SN=Salience Network, SNIG=Substantia Nigra); legacy `brain-nodes-111.json` reintroduces SN=Substantia Nigra. Any registry merge that treats them as one node is a hard error.
- **Case/spelling conflicts (dIPFC/DLPFC, NAc/NAcc, PREC/PRECEUNEUS, DRN/RAPHE): CONFIRMED.**
- **Three "duplicates" are actually subregion-vs-whole lumping, not duplicates:** IFG[⊃]Broca, CA1/CA3/DG[⊃]HIPP, DRN[⊃]RAPHE. Reclassify.
- **Section 3.3 missing-node diff: UNRELIABLE by its own admission** ("needs verification"). Recompute cleanly; do not rely on it.
- **NEW finding — ontological heterogeneity (the audit missed this):** the 123 mix

regions, networks, tracts, molecules, glia, and abstract constructs. Consequences below. This is the blocker for a regulation model.

Non-nodes inside the “node” set (must be handled specially)

Class	Members	Why it can’t take a node’s regulatory role
Tracts (= edges)	CC, FORN, UNC	A tract is a <i>connection between</i> nodes. It has no efferents of its own; it <i>is</i> the efferent. Model as edges.
Molecular systems	GABA_GLU, BDNF, TrkB, OPIOID, OXY	Not localizable. Represent by their source nucleus (e.g. OXY→PVN/HYPO) or as a global modulator parameter.
Glia / barrier	ASTRO, MICRO, BBB	Non-neuronal. Real modulators, but not graph nodes with directed axonal output.
Constructs / states	EI, EMP, DISS, OSC	EI <i>is</i> regulation (a balance parameter); DISS/OSC are system states; EMP is a multi-region construct. None can “regulate others” as a node.
Ambiguous abbreviations	CON, VIA, CMZ, SDH, MI, ENDO	Cannot resolve confidently from the abbreviation alone — flagged inline. Confirm from source.

REGULATORY REFERENCE BY SYSTEM

Format per node: **ABBR** — Name. Definition. *Self*: autoregulation. *Others*: efferent modulation (→ target, sign). [tier/flag]

A. Medial-prefrontal regulatory cortex

- **vmPFC** — Ventromedial PFC. Value, extinction, autonomic top-down control. *Self*: local GABAergic (PV/SST) feedback inhibition. *Others*: → BLA/CeA **inhibitory** (extinction) ✓cited (Milad/Quirk); → HYPO/PAG dampens stress/autonomic output; → NAcc value signals. [SOLID]
- **mPFC** — Medial PFC (dorsal/general). Self-referential processing, DMN hub. *Self*: interneuron E/I balance. *Others*: → HIPPO (memory-guided regulation), → amygdala (context), → DMN coherence. [SOLID]
- **OFC** — Orbitofrontal cortex. Outcome/reward valuation, reversal. *Self*: local inhibition. *Others*: → amygdala, striatum, hypothalamus (value updating). [SOLID]

- **sgACC** — Subgenual ACC (BA25). Mood, autonomic; DMN–sgPFC coupling in depression. *Self*: local inhibition. *Others*: → hypothalamus, brainstem autonomic; hyperactivity ↔ depression ✓cited (Hamilton/DMN set). [SOLID]
- **rACC** — Rostral/pregenual ACC. Emotional appraisal, conflict-emotion interface. *Self*: interneuron balance. *Others*: → amygdala (regulation), dACC. [MODERATE]
- **rPFC** — Rostral/rostrolateral PFC (frontopolar, BA10). Metacognition, abstract integration. *Self*: local inhibition. *Others*: → dlPFC, parietal; metacognitive precision on lower-order judgements ✓cited (Fleming). [MODERATE]
- **MFC** — Medial frontal cortex (general/pre-SMA-adjacent). Action-outcome monitoring. *Self*: local balance. *Others*: → SMA/pre-SMA, dACC. [MODERATE — overlaps mPFC/dACC; check for redundancy]

B. Cingulate

- **ACC** — Anterior cingulate (general). Conflict/error, salience. *Self*: local inhibition. *Others*: → dlPFC (control recruitment), autonomic brainstem. [SOLID]
- **dACC** — Dorsal ACC. Salience/conflict monitoring, part of salience network. *Self*: interneuron balance. *Others*: with AI drives DMN↔ECN switching ✓cited (Sridharan/Seeley); → dlPFC, SMA. [STRONG]
- **CING** — Cingulate cortex (whole/posterior-inclusive). Broad. [AMBIGUOUS — likely redundant with ACC/PCC/dACC; confirm intended referent]
- **CMZ** — [AMBIGUOUS] likely Cingulate Motor Zone. If so: cingulate premotor region, action selection under conflict. *Others*: → M1/SMA/spinal. [CONFIRM FROM SOURCE]

C. Insula

- **AI** — Anterior insula. Interoceptive salience, part of salience network. *Self*: local inhibition. *Others*: with dACC drives network switching ✓cited (Sridharan); → amygdala, ACC. [STRONG]
- **PI** — Posterior insula. Primary interoceptive cortex (bodily afferents). *Self*: local balance. *Others*: → AI (interoceptive integration). [SOLID]
- **MI** — [AMBIGUOUS] likely Mid-insula (interoceptive integration between PI and AI). [CONFIRM FROM SOURCE]

D. Lateral frontal, premotor, motor, eye fields

- **dlPFC** — Dorsolateral PFC. Working memory, top-down cognitive control. *Self*: recurrent PV-interneuron inhibition sustaining persistent activity. *Others*: → parietal, striatum,

sensory cortex — biases processing toward goals ✓cited (Miller & Cohen). [SOLID]

- **vIPFC** — Ventrolateral PFC. Response inhibition, retrieval, reappraisal. *Self*: local inhibition. *Others*: → amygdala (reappraisal-linked down-regulation), basal ganglia (stopping). [SOLID]
- **FPC** — Frontopolar cortex (BA10). Branching, exploration, relational integration. *Self*: local balance. *Others*: → dlPFC. [MODERATE — near-duplicate of rPFC; reconcile]
- **PMC** — Premotor cortex. Movement preparation/selection. *Self*: local inhibition. *Others*: → M1, spinal. [SOLID]
- **SMA** — Supplementary motor area. Internally-generated action, sequencing. *Self*: local balance. *Others*: → M1, basal ganglia loop. [SOLID]
- **M1** — Primary motor cortex. Motor execution. *Self*: Renshaw/interneuron feedback (at spinal target), cortical inhibition. *Others*: → corticospinal → spinal motor neurons. [SOLID]
- **FEF** — Frontal eye fields. Saccade control, spatial attention (dorsal attention). *Self*: local inhibition. *Others*: → SC, parietal (attentional biasing). [SOLID]

E. Parietal

- **S1** — Primary somatosensory. Tactile/proprioceptive map. *Self*: lateral inhibition (sharpening). *Others*: → S2, M1, PI. [SOLID]
- **S2** — Secondary somatosensory. Higher tactile, integration. *Self*: local inhibition. *Others*: → insula, parietal. [SOLID]
- **IPL** — Inferior parietal lobule. Multimodal integration, part of DMN/TPJ region. *Self*: local balance. *Others*: → prefrontal, temporal. [MODERATE]
- **SPL** — Superior parietal lobule. Spatial attention, sensorimotor integration (dorsal attention). *Self*: local inhibition. *Others*: → FEF, premotor. [SOLID]
- **IPS** — Intraparietal sulcus. Attention, number, reach/grasp; dorsal attention hub. *Self*: local balance. *Others*: → FEF (fronto-parietal attention), premotor. [SOLID]
- **AG** — Angular gyrus. Semantic integration, DMN node. *Self*: local balance. *Others*: → DMN nodes, temporal. [MODERATE]

F. Temporal, occipital, visual/auditory cortex

- **A1** — Primary auditory cortex. *Self*: lateral inhibition. *Others*: → auditory association, STS. [SOLID]

- **V1** — Primary visual cortex. *Self*: lateral/surround inhibition; predictive feedback target ✓cited (Rao & Ballard). *Others*: → V2+, dorsal/ventral streams. [SOLID]
- **V4V5** — Extrastriate visual (V4 color/form, V5/MT motion). *Self*: local inhibition. *Others*: → parietal (dorsal), temporal (ventral). [SOLID]
- **FG** — Fusiform gyrus. Face/object/word form. *Self*: local balance. *Others*: → amygdala (face-emotion), temporal. [SOLID]
- **EBA** — Extrastriate body area. Body perception. *Self*: local balance. *Others*: → parietal, premotor. [MODERATE]
- **PPA** — Parahippocampal place area. Scene/context. *Self*: local balance. *Others*: → HIPP (contextual memory), RSC. [SOLID]
- **STS** — Superior temporal sulcus. Biological motion, social perception, gaze. *Self*: local balance. *Others*: → amygdala, TPJ, FG (social pipeline). [SOLID]
- **TPOLE** — Temporal pole. Semantic hub, social-emotional. *Self*: local balance. *Others*: → amygdala, OFC. [MODERATE]
- **PRC** — Perirhinal cortex. Object recognition memory, MTL gateway. *Self*: local inhibition. *Others*: → EC → HIPP. [SOLID]
- **EC** — Entorhinal cortex. HIPP gateway, grid cells. *Self*: local inhibition. *Others*: → HIPP (perforant path), neocortex. [SOLID]
- **RSC** — Retrosplenial cortex. Spatial/contextual memory, DMN-linked. *Self*: local balance. *Others*: → HIPP, PPA, PCC. [SOLID]

G. Language

- **BROCA** — Broca's area (IFG pars). Speech production, syntax. *Self*: local inhibition. *Others*: → motor speech, WERN (arcuate). [SOLID] (*Note: IFG → Broca — see lumping flag.*)
- **WERN** — Wernicke's area. Language comprehension. *Self*: local balance. *Others*: → BROCA, semantic temporal. [SOLID]
- **LANG** — Language network (composite). [COMPOSITE — fronto-temporal language system; do not also edge as atomic. Read-only computed view.]

H. Large-scale networks (COMPOSITES — flag as read-only computed views, not edge endpoints)

- **DMN** — Default mode network. Self-referential/internal mentation. *Regulation*:

anticorrelated with task-positive; quieted by salience switching ✓cited (Sridharan, Brewer). [COMPOSITE]

- **DMNMTL** — DMN medial-temporal subsystem. Memory-based simulation. [COMPOSITE — subsystem of DMN; double-count risk with HIP/DMN.]
- **SN** — Salience network (AI+dACC). Detects salient events, switches DMN↔ECN ✓cited (Seeley/Sridharan). [COMPOSITE] ⚠ **abbreviation collides with Substantia Nigra — keep SNIG distinct.**
- **FPN / FPC** — Frontoparietal (control) network. Flexible task control. [COMPOSITE — reconcile FPN vs FPC vs ECN; likely overlapping labels.]
- **ECN** — Executive control network. Goal-directed control. [COMPOSITE — near-synonym of FPN; dedupe.]
- **DAN** — Dorsal attention network. Voluntary spatial attention (FEF+IPS). [COMPOSITE]
- **VAN** — Ventral attention network. Stimulus-driven reorienting. [COMPOSITE]
- **CON** — [AMBIGUOUS] likely Cingulo-Opercular Network (tonic control/alertness). [CONFIRM — if so, COMPOSITE, overlaps dACC/AI.]

I. Medial-temporal / limbic

- **BLA** — Basolateral amygdala. Emotional learning, fear acquisition. *Self*: intercalated + local GABAergic feedback inhibition. *Others*: → CeA (fear output), → NAcc, mPFC; gated **inhibitorily** by vmPFC ✓cited. [SOLID]
- **CeA** — Central amygdala. Fear output hub. *Self*: internal GABAergic microcircuits (CeL→CeM inhibition). *Others*: → PAG (freezing), HYPO (HPA/autonomic), brainstem. [SOLID]
- **BNST** — Bed nucleus stria terminalis. Sustained anxiety, threat monitoring. *Self*: internal GABAergic. *Others*: → HYPO (HPA), brainstem autonomic. [SOLID]
- **HIPP** — Hippocampus. Episodic/spatial memory, HPA feedback. *Self*: strong GABAergic interneuron feedback (theta/gamma). *Others*: → mPFC, EC; → HYPO **inhibitory** (glucocorticoid negative feedback). [SOLID] (*CA1/CA3/DG are subfields — see lumping flag.*)
- **HAB** — Habenula (lateral). Aversion, "anti-reward," negative prediction. *Self*: local balance. *Others*: → RAPHE/VTA **inhibitory** (suppresses reward/serotonin). [SOLID]
- **SEPT** — Septum (medial/lateral). Theta pacing, social/emotional. *Self*: local balance. *Others*: → HIPP (theta), HYPO. [MODERATE]

- **MAMM** — Mammillary bodies. Memory circuit (Papez). *Self*: relay. *Others*: → ANT thalamus (mammillothalamic). [MODERATE]
- **FORN** — Fornix. [TRACT — HIPPO ↔ MAMM/septum output pathway. Model as edge, not node.]
- **UNC** — Uncinate fasciculus (likely). [TRACT — OFC/vmPFC ↔ anterior temporal/amygdala. Model as edge.] [CONFIRM: uncinatus vs uncus]

J. Hypothalamus & neuroendocrine control

- **HYPO** — Hypothalamus. Homeostatic master (autonomic, endocrine, drive). *Self*: internal nuclei cross-inhibition; glucocorticoid negative feedback. *Others*: → PIT (releasing hormones), autonomic brainstem, → whole-body via HPA. [SOLID]
- **HPA** — Hypothalamic-pituitary-adrenal axis (composite). Stress endocrine cascade. *Self*: **cortisol negative feedback** onto HYPO/PIT/HIPP. *Others*: systemic cortisol. [COMPOSITE/AXIS — represent as HYPO → PIT → ADR loop, not a point node.]
- **PIT** — Pituitary. Endocrine relay. *Self*: feedback from target hormones. *Others*: → ADR (ACTH), thyroid, gonads. [SOLID]
- **ADR** — Adrenal (gland). Cortisol + catecholamine output. *Self*: feedback via cortisol on axis. *Others*: systemic (cortisol, epinephrine). [PERIPHERAL organ]
- **PIN** — Pineal. Melatonin/circadian. *Self*: driven by SCN. *Others*: systemic melatonin. [PERIPHERAL]
- **SCN** — Suprachiasmatic nucleus. Master circadian clock. *Self*: molecular clock autoregulation (transcription-translation loop). *Others*: → PIN, HYPO, autonomic (circadian gating). [SOLID]

K. Basal ganglia

- **CAUD** — Caudate. Goal-directed/associative loop. *Self*: local GABAergic. *Others*: → GP/SNIG (→ THAL). [SOLID]
- **PUT** — Putamen. Sensorimotor/habit loop. *Self*: local GABAergic. *Others*: → GP → THAL (basal-ganglia gate). [SOLID]
- **STRI** — Striatum (composite of CAUD+PUT+NAcc). [COMPOSITE — double-count risk with its parts.]
- **NAcc** — Nucleus accumbens (ventral striatum). Reward, motivation, limbic-motor interface. *Self*: local GABAergic; DA modulation. *Others*: → VP → THAL; gates motivated action. [SOLID]

- **GP** — Globus pallidus (internal/external). BG output/relay. *Self*: GABAergic. *Others*: → THAL **inhibitory** (GPi), → STN (GPe); core of the BG gate. [SOLID]
- **VP** — Ventral pallidum. Limbic BG output. *Self*: GABAergic. *Others*: → MDT thalamus, brainstem. [SOLID]
- **STN** — Subthalamic nucleus. Hyperdirect “brake,” global inhibition. *Self*: reciprocal with GPe. *Others*: → GPi **excitatory** (raises inhibitory tone → stopping). [SOLID]
- **SNIG** — Substantia nigra (pars compacta DA + pars reticulata output). *Self*: D2 autoreceptor feedback (SNc). *Others*: SNc → striatum **dopaminergic**; SNr → THAL/SC **inhibitory**. [SOLID] **△ distinct from SN=Salience**.

L. Thalamus

- **THAL** — Thalamus (whole). Sensory/cortical relay, gating, arousal. *Self*: **thalamic reticular nucleus** feedback inhibition (spindles/gating). *Others*: → cortex (relay), cortico-thalamo-cortical loops. [SOLID] (*Note: TRN not separately in set — the self-regulation mechanism has no node; consider adding.*)
- **ANT** — Anterior thalamic nuclei. Memory (Papez), spatial. *Self*: TRN inhibition. *Others*: → cingulate, HIPPO. [SOLID]
- **MDT** — Mediodorsal thalamus. PFC relay, cognition. *Self*: TRN inhibition. *Others*: → PFC (sustains WM). [SOLID]
- **PULV** — Pulvinar. Visual attention, cortico-cortical coordination. *Self*: TRN inhibition. *Others*: → visual/parietal cortex (attentional synchrony). [SOLID]
- **MGN** — Medial geniculate. Auditory relay. *Self*: TRN. *Others*: → A1. [SOLID]
- **LGN** — Lateral geniculate. Visual relay. *Self*: TRN; strong corticothalamic feedback. *Others*: → V1. [SOLID]

M. Midbrain / brainstem / neuromodulatory nuclei

- **VTA** — Ventral tegmental area. Dopamine source (reward/motivation/salience). *Self*: **D2 autoreceptors** + local GABA interneuron feedback. *Others*: → NAcc, PFC, amygdala **dopaminergic**. [SOLID]
- **LC** — Locus coeruleus. Norepinephrine source (arousal, gain, alerting). *Self*: **α2-adrenergic autoreceptor** negative feedback. *Others*: → near-global cortex/amygdala/thalamus **noradrenergic** (gain/precision broadcast). [SOLID]
- **RAPHE** — Raphe nuclei (dorsal/median). Serotonin source (mood, patience, plasticity). *Self*: **5-HT1A somatodendritic autoreceptors**. *Others*: → near-global **serotonergic**;

suppressed by HAB. [SOLID] (*DRN_cRAPHE* — *lumping flag.*)

- **NBM** — Nucleus basalis of Meynert. Acetylcholine source (cortical). *Self*: local feedback. *Others*: → neocortex **cholinergic** (attention/plasticity gain). [SOLID]
- **PPN** — Pedunculopontine nucleus. Cholinergic/arousal, gait, RAS. *Self*: local balance. *Others*: → THAL, BG, brainstem (arousal). [SOLID]
- **PAG** — Periaqueductal gray. Defensive behavior, descending pain control. *Self*: internal columns. *Others*: → medulla (freezing/analgesia), autonomic; receives CeA/vmPFC. [SOLID]
- **PBN** — Parabrachial nucleus. Interoceptive/visceral relay, arousal, pain/alarm. *Self*: local balance. *Others*: → amygdala, HYPO, THAL, insula. [SOLID]
- **SC** — Superior colliculus. Orienting, saccades, threat detection. *Self*: lateral inhibition. *Others*: → brainstem gaze, PULV. [SOLID]
- **IC** — Inferior colliculus (likely, given A1/MGN present). Auditory midbrain. *Self*: local inhibition. *Others*: → MGN → A1. [CONFIRM: inferior colliculus vs internal capsule(tract) vs insular cortex]
- **RF** — Reticular formation. Arousal, autonomic pattern generation. *Self*: distributed. *Others*: → THAL/cortex (arousal), spinal, autonomic. [SOLID]
- **NTS** — Nucleus tractus solitarius. Primary visceral afferent hub (vagal sensory). *Self*: local GABA. *Others*: → PBN, HYPO, DV/DMV, RVLM; vagal afferent gateway ✓cited-adjacent (Thayer autonomic). [SOLID]

N. Cerebellum

- **CBLM** — Cerebellum (whole). Prediction/timing/error correction (motor + cognitive/affective). *Self*: Purkinje GABAergic output; climbing-fiber error. *Others*: → THAL → cortex; → brainstem. [SOLID]
- **VERM** — Vermis. "Limbic cerebellum," affect/autonomic. *Self*: Purkinje inhibition. *Others*: → fastigial → brainstem/HYPO. [MODERATE]
- **NEOCER** — Neocerebellum (lateral hemispheres). Cognitive/predictive. *Self*: Purkinje inhibition. *Others*: → dentate → THAL → PFC. [MODERATE]

O. Autonomic / vagal / interoceptive periphery

- **DV** — Dorsal vagal complex (DMV/dorsal motor nucleus). Parasympathetic, immobilization. *Self*: NTS feedback. *Others*: → viscera (parasympathetic). [SOLID as anatomy; **polyvagal interpretation CONTESTED** ✓flagged (Porges)]

- **VV** — Ventral vagal complex (nucleus ambiguus). Parasympathetic, social engagement, cardiac. *Self*: NTS feedback. *Others*: → heart/larynx/pharynx. [SOLID as anatomy; **polyvagal framing CONTESTED**]
- **LAR** — Laryngeal (vagal laryngeal control). Vocalization, prosody. *Self*: brainstem feedback. *Others*: → larynx. [MODERATE — peripheral effector]
- **CARD** — Cardiac (control/plexus). Heart-rate autonomic node. *Self*: baroreflex. *Others*: HRV output; indexes vmPFC top-down tone ✓cited (Thayer). [PERIPHERAL/effector]
- **ENS** — Enteric nervous system. Gut "second brain." *Self*: intrinsic reflex loops. *Others*: → vagal afferents (NTS), gut-brain. [SOLID]
- **GBA** — Gut-brain axis (composite). *Others*: microbiome/vagal/immune → brain. [COMPOSITE/AXIS — not a point node. GUT is a duplicate alias.]
- **SNS** — Sympathetic nervous system. Fight/flight autonomic output. *Self*: baroreflex/central feedback. *Others*: → viscera/adrenal (norepinephrine/epinephrine). [SYSTEM/effector]
- **SDH** — [AMBIGUOUS] likely Spinal Dorsal Horn (nociceptive gate; gate-control, descending PAG modulation). [CONFIRM FROM SOURCE]

P. Association / other cortex

- **PCC** — Posterior cingulate. DMN core, self/internal awareness; reconfigures in meditation. *Self*: local balance. *Others*: → DMN nodes; PCC↔DLPFC coupling in meditators ✓cited (Brewer). [SOLID]
- **PRECUNEUS** — Precuneus. Self-processing, imagery, DMN. *Self*: local balance. *Others*: → PCC, parietal, DMN. [SOLID] (*PREC is generator drift — same node.*)
- **TPJ** — Temporoparietal junction. Theory-of-mind, reorienting, agency. *Self*: local balance. *Others*: → mPFC (social), VAN. [SOLID]

Q. Tracts — MODEL AS EDGES, NOT NODES

- **CC** — Corpus callosum. Interhemispheric tract. [EDGE — connects homotopic cortex L↔R.]
- **FORN** — Fornix. [EDGE — HIPPO output.]
- **UNC** — Uncinate fasciculus. [EDGE — OFC/vmPFC↔temporal/amygdala.]

R. Molecular / neurochemical systems — NOT REGIONAL NODES

- **GABA_GLU** — GABA/glutamate (inhibition/excitation transmitters). [PARAMETER — this

is the E/I substrate; represent as edge sign + gain, not a node.]

- **BDNF** — Brain-derived neurotrophic factor. Plasticity signal. [MOLECULE — model as a plasticity/gain modulator, not a node.]
- **TrkB** — BDNF receptor. [MOLECULE — receptor for BDNF; not a node.]
- **OPIOID** — Endogenous opioid system. Analgesia, reward, affiliation. [MODULATOR — sources include PAG, ARC; not a point node.]
- **OXY** — Oxytocin. Bonding, trust, co-regulation. [NEUROPEPTIDE — source = PVN/HYPO; represent as HYPO efferent modulator.]

S. Glia & barrier — NON-NEURONAL

- **ASTRO** — Astrocytes. Glutamate uptake, gliotransmission, metabolic support, tripartite synapse. [GLIAL MODULATOR — not a directed node.]
- **MICRO** — Microglia. Immune surveillance, synaptic pruning. [GLIAL/IMMUNE — not a directed node.]
- **BBB** — Blood-brain barrier. Vascular interface. [STRUCTURE — not a node.]

T. Constructs / states / unresolved — CANNOT BE NODES AS-IS

- **EI** — Excitation/Inhibition balance. [PARAMETER — *is* regulation itself, not a regulated node. Model as edge-sign balance.]
- **OSC** — Oscillation. [STATE/PROPERTY — emergent dynamic (theta/gamma), not a node.]
- **EMP** — Empathy. [CONSTRUCT — multi-region (AI, ACC, TPJ, STS). Represent as computed view, not atomic node.]
- **DISS** — Dissolution. [STATE — LIMEN Phase-7 construct / entropic state, not an anatomical node. Do not give efferents.]
- **ARC** — Arcuate nucleus (likely). Hypothalamic feeding/endocrine, β -endorphin. *Others:* → HYPO, PIT. [CONFIRM: arcuate nucleus vs arcuate fasciculus(tract)]
- **CON** — [AMBIGUOUS — cingulo-opercular network? confirm]
- **VIA** — [AMBIGUOUS — cannot resolve; confirm from source]
- **CMZ** — [AMBIGUOUS — cingulate motor zone? confirm]
- **MI** — [AMBIGUOUS — mid-insula? confirm]

- **ENDO** — [AMBIGUOUS — endocrine general? or endocannabinoid system? confirm]
 - **EI, OSC, DISS, EMP** — see above.
-

WHAT THIS BLOCKS AND WHAT'S NEXT

1. **Sort the set before modeling regulation.** ~85 real nodes can carry self/other regulation. The ~38 others (composites, tracts, molecules, glia, constructs, ambiguous) cannot as-is and must be reclassified: composites→read-only views, tracts→edges, molecules/E-I→edge parameters, constructs→computed views, ambiguous→disambiguate from source.
2. **Disambiguate the flagged abbreviations** (CON, VIA, CMZ, SDH, MI, ENDO, ARC, IC, UNC) against the source docx/JSON before they get regulatory roles.
3. **TRN is missing.** The self-regulation mechanism for the entire thalamus (reticular-nucleus feedback inhibition) has no node. If thalamic gating matters to the model, add it.
4. **Next pass = reconcile "regulates others" against the actual L2 edges.** This document is the literature-predicted edge set with signs. Diff it against `brain-connectome.json`'s 65 encoded edges to find: which regulatory edges exist, which are missing, and which have the wrong/absent sign. That diff is the concrete spec for the inhibition-first fix.
5. **Optional:** DOI-verify the specific regulatory mechanisms (autoreceptors, vmPFC→amygdala sign, HIPPO→HYPO feedback) if any are going to render as cited claims rather than textbook background.

Regulatory relationships are literature-standard neuroanatomy stated as consensus, not re-verified DOIs this session (except items marked ✓cited). Verified-real ≠ established-true remains the standard.